

1988 IMO 6

Show that for positive integers a and b , if $\frac{a^2+b^2}{ab+1}$ is an integer, then it is a perfect square.

$$\frac{a^2+b^2}{ab+1} = k \quad (k > 0)$$

$$a^2 + b^2 = abk + k$$

$$a^2 - bka + b^2 - k = 0$$

If (a, b) is a solution, then,

$\exists$ a solution a' such that
 $a + a' = bk$ and $aa' = b^2 - k$.
 $a' = bk - a = \frac{b^2 - k}{a}$.

If (a, b) is a solution, then
 $(bk - a, b)$ is also a solution.

$$a' = \frac{b^2 - k}{a} < \frac{b^2}{a} < \frac{a^2}{a} = a \quad \therefore a^2 < a.$$

$$a'^2 + b^2 = a'b + k$$

$$\text{FTIC} \quad a' < 0$$

$$a'^2 + b^2 > 0$$

$$a'b + k = k(a'b + 1) \leq k(-b + 1) \leq 0$$

$$a' = 0$$

$$b^2 = k$$

$$\therefore a' > 0$$

By well-ordering principle,
 we can let a solution
 (a, b) such that $a+b$ is
 least.

FTIC, assume that
 k is not a perfect
 square.
 WLOG, $a > b$.

It suffices to show
 that $a' + b < a + b$
 and $a' > 0$.

□